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Prograding Carbonate Ramp Environments and its Implications on the Reservoir Distribution in the Minagish Formation of Greater Burgan field

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Abstract

This study aims to examine the prograding carbonate ramp environments of the Minagish Formation in the Greater Burgan Field and their implications for reservoir distribution. The study focuses on understanding the transition of depositional environments within the Minagish formation utilizing regional Sedimentological, Petrographic, well log data, core descriptions, and facies interpretations. The key objective of this study is to refine the depositional model by addressing uncertainties in facies correlations and assessing the implications on the reservoir quality and connectivity of Minagish reservoir of Burgan field across adjacent fields in Kuwait. Insights from core data and well logs assist in enhancing reservoir management strategies. The study utilizes a multi-disciplinary approach combining regional well Sedimentology, Petrography, log analysis, core descriptions, and facies interpretations to investigate the depositional environments of the Minagish Formation. Well log data and core samples from various fields are analyzed to identify facies transitions and their implications for reservoir distribution. Facies models are constructed based on sedimentological observations, to refine the understanding of the prograding carbonate ramp. The methodology includes detailed study of sedimentological data, well log correlation, facies mapping, and comparison with existing models to address uncertainties and improve regional reservoir characterizations, ultimately aiding in reservoir management and decision-making. The study of progradational carbonate ramp environments in the Minagish Formation offers critical insights into the depositional trends and facies variability across multiple fields, including Umm Gudair, Minagish, Raudhatain, Sabriyah, Rugei, and others. Observations indicate a shift from lagoonal to shoal facies in the Umm Gudair Field. Core data from various fields suggest a predominance of mid-outer ramp facies in the north and northeast, while the Minagish field shows a progression from outer ramp to shoal and lagoonal facies. The study underscores the complexities of carbonate ramp systems, highlighting dynamic depositional environments influenced by both high and low-energy conditions, with episodic storm events evident in some fields. The findings support the regional depositional models for the Burgan-Minagish field, offering refined insights into the spatial distribution of reservoir facies and their implications for reservoir management and field development. This paper contributes new insights by refining the understanding of the prograding carbonate ramp environments in the Minagish Formation. By analyzing

regional sedimentological, petrographic study, well logs, core descriptions, and facies interpretations, it provides a clearer picture of the distribution of reservoir facies across Kuwait. The study strengthens the regional correlation of carbonate ramp systems, identifies variations in depositional settings, and confirms the dynamics of carbonate ramp progression. These findings help improve reservoir management and reduce uncertainties associated with subsurface facies distribution.

Introduction

The Burgan Minagish Oolite reservoir of Barresian-Valanginian age was discovered in 1950. It is one of four main conventional reservoirs in the Greater Burgan field. The Burgan Minagish reservoir is a four-way anticline with bottom water reservoir drive mechanism. There is limited faulting within the reservoir and there is no evidence of significant compartmentalization within the reservoir. The reservoir structure is underfilled, with evidence of structural movement and up-dip remigration of hydrocarbons resulting in the presence of paleo-contact and the occurrence of residual oil-saturations below the present-day original oil water contact (OOWC). Minagish reservoir is hydrocarbon bearing only in the crest of Burgan field where high-quality reservoir rocks are present. The reservoir quality degrades into non-reservoir basinal facies towards the northern part in the Magwa and Ahmadi areas of the Greater Burgan Field. The reservoir was shut-in from 2005-2009 due to high H₂S content, during this period, there was a significant drop in the reservoir pressure even though all the wells were shut in which was attributed to the off take in adjacent West Kuwait fields by previous reservoir studies. This study focuses on understanding the transition of depositional environments within the Minagish formation utilizing regional sedimentological, petrographic, well log data, core descriptions, and facies interpretations. The key objective of this study is to refine the depositional model by addressing uncertainties in facies correlations and assessing the implications on the reservoir quality and connectivity of Minagish reservoir of Burgan field across adjacent fields in Kuwait.

The study utilizes a multi-disciplinary approach combining regional well sedimentology, petrography, log analysis, core descriptions, and facies interpretations to investigate the depositional environments of the Minagish Formation. Well log data and core samples from various fields are analyzed to identify facies transitions and their implications for reservoir distribution. Facies models are constructed based on sedimentological observations, to refine the understanding of the prograding carbonate ramp. The methodology includes detailed study of sedimentological data, well log correlation, facies mapping, and comparison with existing models to address uncertainties and improve regional reservoir characterizations, ultimately aiding in reservoir management and decision-making.

Geological Setting

The Inner Ramp Shallow Subtidal (clay-prone) association in the upper Minagish consists of claystone marls, mudstones, and argillaceous packstones, marked by high gamma ray readings and low porosity (Fig.1). It forms during transgressive phases in low-energy conditions, with limited faunal content and poor reservoir quality. Microbial Inner Ramp Lagoon deposits are rare but host moderate to good reservoir quality with porosity between 15-30%. These deposits reflect low-energy, episodic conditions, often disturbed by storms. The oncolitic, rudist-prone, and skeletal-peloidal shoal associations, typically found in the middle Minagish, are characterized by oncoidal, rudist, and skeletal-peloidal grainstones, reflecting deposition in shallow, current-agitated environments. The algal shoal association, the most abundant, is known for its high reservoir quality with porosity of 15-30% and permeability ranging from 10-8000mD. The oolitic shoal complex, found mainly in the lower Minagish, consists of oolitic grainstones and pack-grainstones, indicating high-energy environments with good reservoir quality. Mid-ramp deposits, interpreted from wireline logs, are characterized by bioturbated skeletal-peloidal packstones, deposited below fair-weather wave base in deeper water. These facies reflect a progradational sequence, with shifts from high-energy inner ramp to more offshore settings.

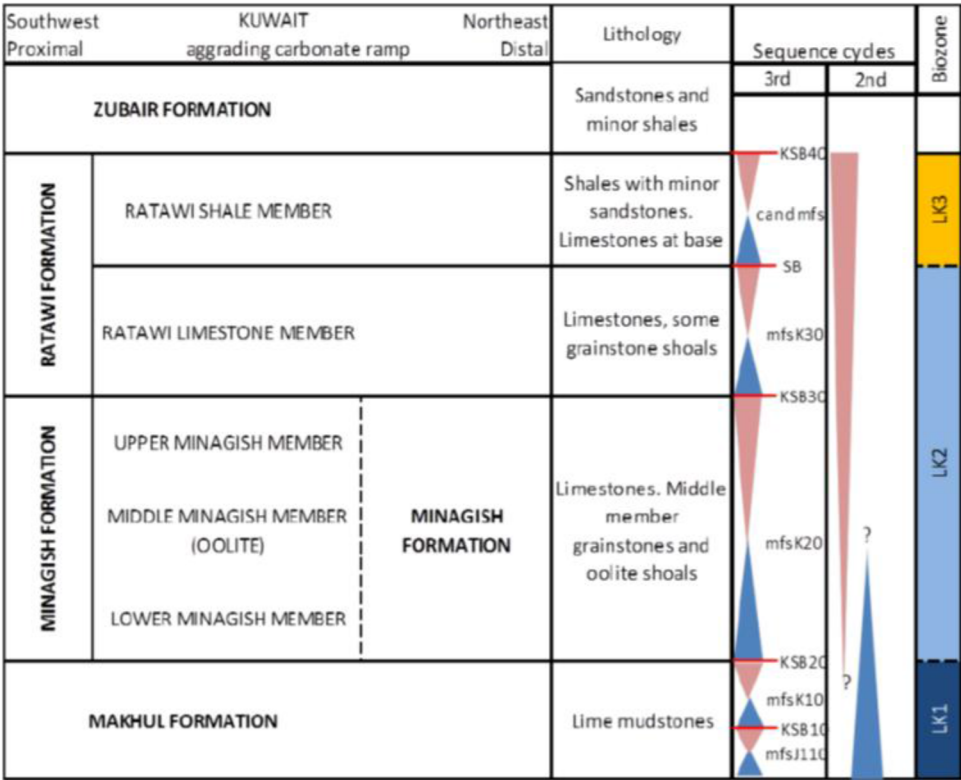


Figure 1—Summary of Stratigraphic column (After Crittenden et al., 2012)

Carbonate depositional environments are diverse and are typically arranged along a ramp profile from the shoreline to the deep basin, reflecting variations in energy, water depth, and biological activity. On the inner ramp, adjacent to the continental margin, environments such as restricted tidal flats and lagoons are common, often characterized by low energy conditions and the deposition of fine-grained sediments (Fig.2).

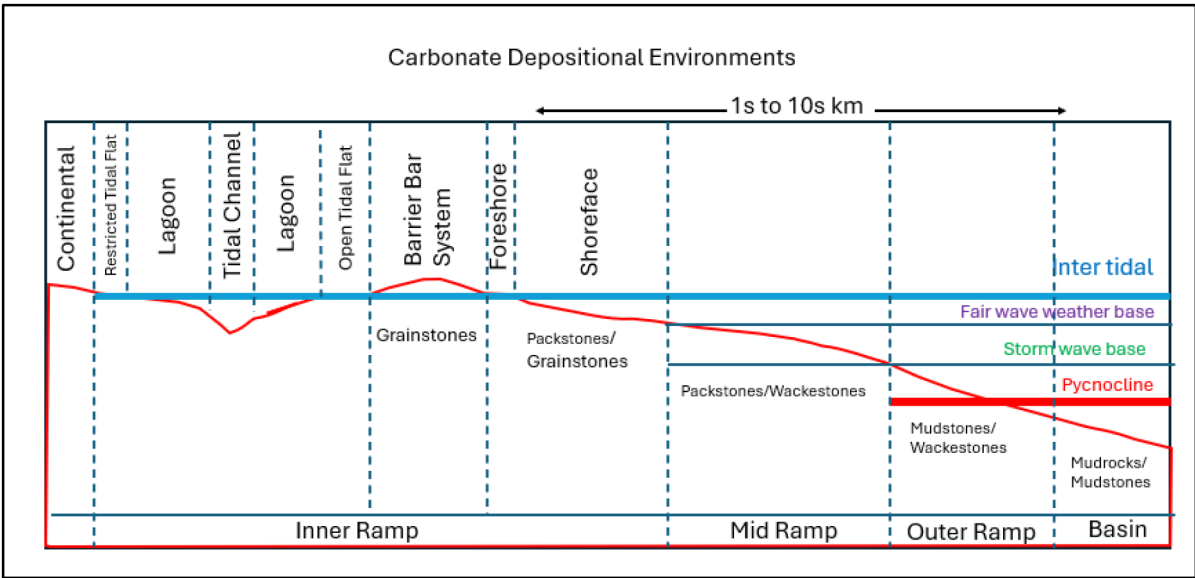


Figure 2—showing carbonate Depositional Environments and the Facies Associations After Green et al. (2018).

Tidal channels within this zone may contain coarser materials due to periodic strong currents, while open tidal flats and barrier bar systems develop in more exposed settings, commonly associated with grainstone

lithofacies due to higher wave and tidal energy. Foreshore and shoreface zones along the barrier may host well-sorted packstones and grainstones, deposited under moderate to high energy conditions. Moving offshore, the mid ramp is a transitional zone with reduced energy, where packstones and wackestones dominate, reflecting a mix of biological and physical sedimentation. Further basinward, on the outer ramp, energy levels are low and the dominant lithofacies are mudstones and wackestones, deposited in relatively deeper, quieter water. Finally, in the basinal setting, the deepest part of the system, fine-grained mudrocks and mudstones accumulate, reflecting suspension settling in low-energy, anoxic conditions often far from carbonate-producing sources. This gradation in facies from inner to outer ramp highlights the interplay between energy, water depth, and sediment supply in shaping carbonate platforms.

Depositional Model of Burgan Minagish reservoir

The depositional model of the Minagish reservoir in the Burgan Field, as derived from an extensive well dataset and sequence stratigraphic analysis, provides a detailed framework for understanding the lateral and vertical heterogeneity of the platform's depositional environments (Fig. 3). The lower Minagish is characterized by a progradational oolite-rimmed carbonate ramp, with evidence pointing to a northeastward progradation of the platform. The gamma ray logs and the cored oolite occurrences confirm a progradational shoal system, with high lateral facies continuity in offshore mid-ramp and foreshoal settings. As the system progrades into oolite shoals, lateral continuity decreases, and thick oolitic intervals form northeastward shoaling geobodies, stacked in progradational units. Oolitic shoals act as barriers, creating protected high-energy backshoal and lagoonal environments to the southwest. These facies are interbedded with peloidal-skeletal shoals and wash-over units, which indicate continued regression and platform progradation.

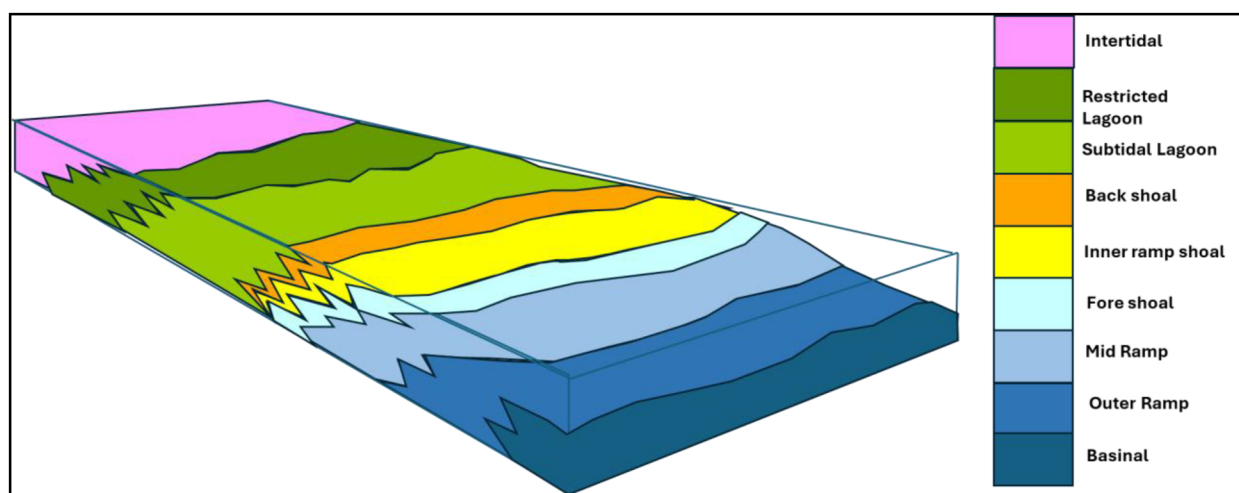


Figure 3—Depositional Model of Burgan Minagish reservoir

The middle Minagish, representing a platform top with backshoal to lagoonal environments, developed behind the northeastward migrating oolitic shoal complex (Fig. 4). Oolitic shoal facies are locally developed in the northeastern part of the Burgan Field, with more complex depositional settings, including algal, rudist, and skeletal shoals, as well as oncoidal and microbial facies. These facies are interbedded with backshoal deposits, revealing significant lateral facies heterogeneity suggesting complex ecological controls driven by tidal currents and minor topographic variability. Increased abundances of *Lithocodium/Bacinella* and algal grains dominate the northeastern wells, while oncotic and rudist-dominated facies are more prominent in the southwestern wells, indicating a more proximal, landward setting with reduced accommodation space and slightly lower depositional energy.

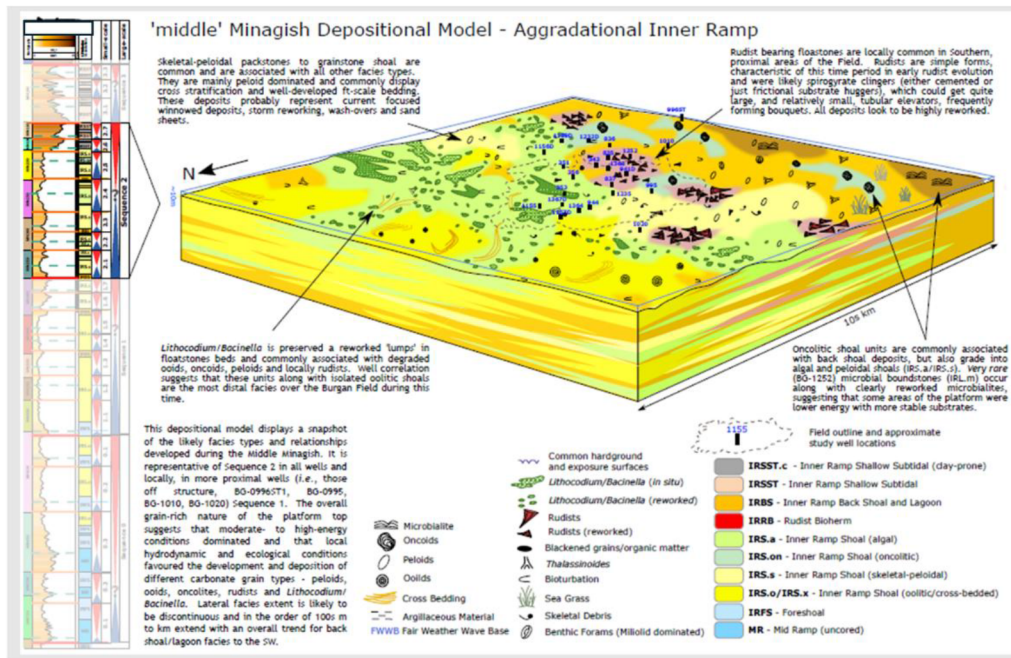


Figure 4—Depositional model of the 'middle' Minagish, Burgan Field. Green et al. (2018).

In the upper Minagish, a distinct shift in depositional style occurs, marked by a progressive decrease in carbonate production due to increased clay input and transgression (Fig 5). This transition is attributed to the influx of clastic material, likely from the proximity of fluvial systems, which suppressed carbonate deposition. The dilute clastic supply in this phase is seen as a precursor to the more significant fluvio-deltaic systems that characterize later deltaic pulses in southwestern Kuwait. Shallow coastal environments, evidenced by well-developed *Thalassinoides* burrow networks and firmgrounds, dominate early deposition. The alternating clay and carbonate layers in the upper Minagish reflect repeated flooding events that reduced the clay supply, promoting cleaner carbonate deposition in shallow waters. Throughout this section, the depositional energy continued to decrease, with grain size reducing progressively. The model suggests a northwest-dipping, prograding homoclinal ramp, with a dominance of shallow subtidal and clay-prone shallow subtidal settings. Lateral facies continuity is high in this setting, creating a "layer-cake" stacking pattern with relatively low lateral variability compared to the middle Minagish. Overall, the depositional history of the Minagish reservoir reveals a dynamic platform system, with lateral facies variability influenced by both ecological conditions and eustatic changes, leading to variations in reservoir quality across the Burgan Field.

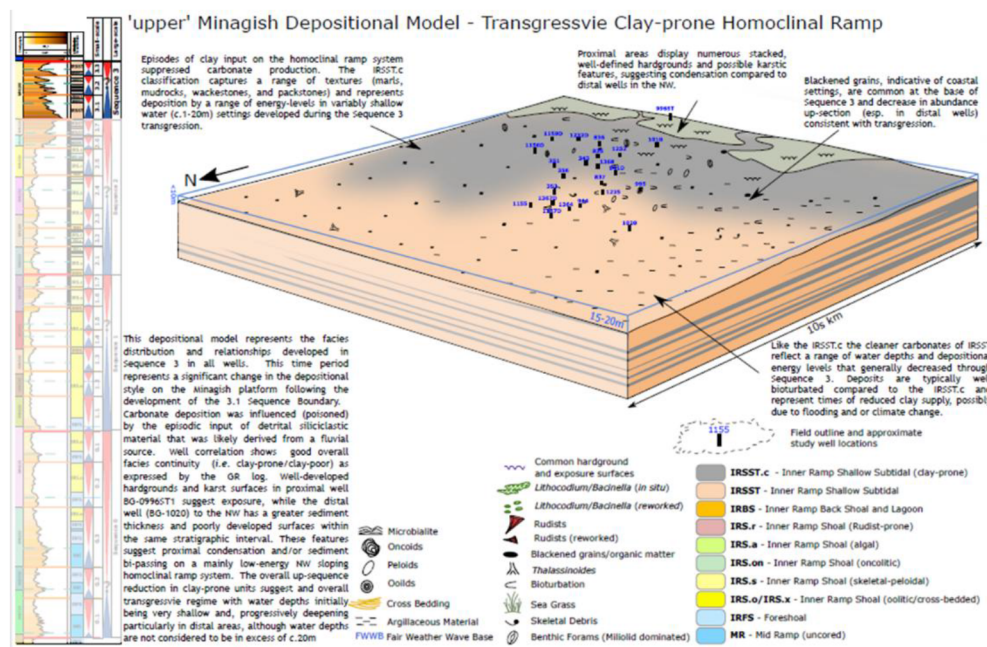


Figure 5—Depositional model of the 'Upper' Minagish, Burgan Field. Green et al. (2018).

Minagish Formation reservoir quality distribution

Sedimentological and Petrographic analysis shows that the sediments are mainly grain-supported, ranging from fine to coarse grains, and were deposited in a shoal complex overlaid by a clay-influenced subtidal lagoon. This results in a linear distribution of porosity and permeability values, with a fracture trend seen in samples with porosity under 3%. Most data clusters exhibit porosity values between 10-35% and permeability between 5-8000 mD, indicating high reservoir quality. The formation predominantly consists of floatstones and rudstones, along with some grainstones, pack-grainstones, and packstones. Matrix-rich textures, such as wackestone and mudstone, exhibit lower porosity and permeability (<15%, <10 mD). Core plugs taken from both horizontal and vertical orientations show slightly better porosity and permeability in horizontal plugs for heterogeneous fabrics, though the difference is minimal in homogeneous fabrics. Reservoir quality is influenced by a combination of depositional settings and diagenetic processes, including texture, hydrodynamic energy, water depth, and grain type. These factors control primary pore architecture and the sorting, packing, and roundness of grains. Diagenetic processes such as compaction, cementation, and dissolution further modify reservoir quality. Compaction reduces pore space, but early marine cementation helps mitigate this. Pressure solution in clay-rich textures can lead to stylolite formation, while dissolution, particularly of susceptible grains, enhances pore volumes, contributing significantly to reservoir quality.

Factors affecting reservoir quality

The distribution of reservoir quality in the Minagish Formation is influenced by a variety of factors, including texture, depositional setting, grain type, compaction, cementation, and dissolution. Grain-supported textures tend to result in better porosity and permeability due to more interconnected pores, while matrix-supported textures, which contain finer materials, often lead to lower reservoir quality (Fig. 6). The depositional setting, influenced by hydrodynamic energy and water depth, affects the size, sorting, and shape of grains, with finer particles like clays reducing porosity and permeability. The type of grains, influenced by the depositional environment and source areas, controls packing and pore architecture, which in turn affects diagenetic processes and long-term reservoir quality. Compaction reduces pore spaces by packing grains closer together, but this can be mitigated by early cementation. Over time, pressure solution replaces compaction, generating stylolites and modifying packstone formation. Cementation in the

Minagish Formation is relatively low, mainly influenced by localized pressure solution processes rather than large-scale flushing. Dissolution, the primary mechanism for pore generation, enhances porosity by dissolving susceptible grains and improving pore connectivity. The best reservoir quality is observed in pack-grainstones and grainstones, where micropore connectivity is improved, allowing for higher fluid flow rates

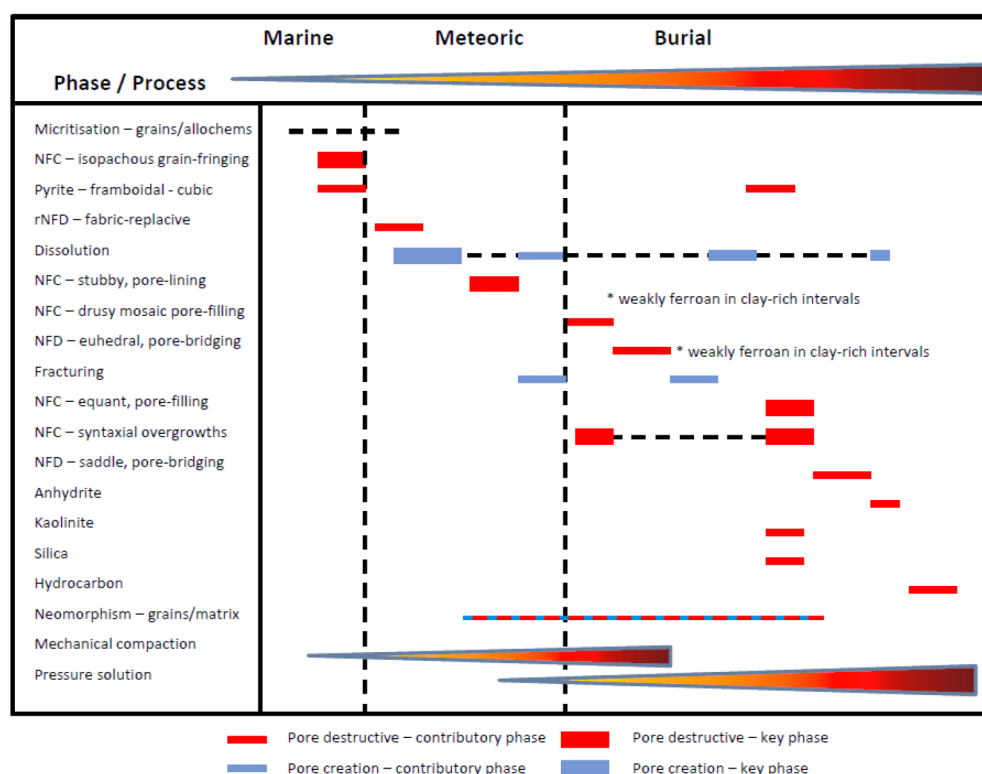


Figure 6—Paragenetic sequence for the cored interval of the Minagish Formation in Burgan Field. Green et al. (2018).

Depositional Effects

Depositional setting plays a critical role in determining reservoir quality in the Minagish Formation. For grain-rich textures, such as those found in Shoal-related settings, the porosity and permeability data exhibit a broad distribution across the positive normal trend. The matrix composition of these rocks, particularly in floatstones and pack-grainstones, significantly influences reservoir properties, with a general tendency for higher permeability and porosity values in these facies. The backshoal settings show a similar distribution, but with a concentration of data points in the upper range of good reservoir properties, reflecting the presence of both grain-rich and matrix-lean fabrics that enhance fluid flow (Fig. 7). Conversely, matrix-rich textures, commonly found in clay-rich lagoonal settings, exhibit depressed porosity and permeability values due to low primary pore volumes and the effects of cementation and compaction, resulting in limited or poorly connected pore spaces. These textures, particularly in claystones and marls, show a trend toward low reservoir potential. Furthermore, depositional factors such as hydrodynamic energy, water depth, and grain sorting also impact the overall pore architecture. While grain size analysis reveals weak trends in permeability, with finer grains clustering toward the poorer reservoir quality end, coarser grains tend to correlate with higher reservoir potential. Additionally, the presence of *Thalassinoides* burrows in lagoonal settings indicates localized variations in porosity and permeability, though these trends are influenced by both depositional and diagenetic processes. The combination of these depositional and diagenetic factors ultimately governs the distribution of reservoir quality throughout the Minagish Formation.

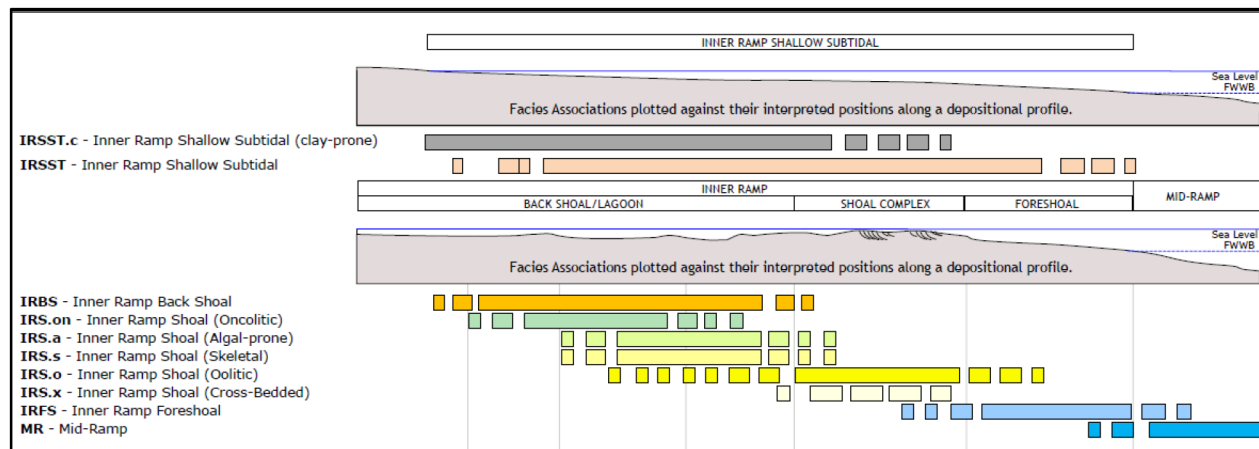


Figure 7—Positions of Lithofacies Associations along the depositional profile. Green et al. (2018).

Diagenetic Effects

The depositional setting has a significant impact on reservoir quality through its influence on pore architecture and diagenetic processes. Grain-rich fabrics, such as those found in shoal and backshoal settings, tend to have better permeability and porosity due to the development of interparticle porosity. These fabrics, particularly pack-grainstones and floatstones, serve as permeability fairways during burial. Interparticle cements, primarily composed of non-ferroan calcite, are more common in permeable grain-rich fabrics and are most abundant in pack-grainstones and floatstones, where they help to improve fluid flow. However, these cements are generally present in low volumes, and their presence can vary across different facies. In contrast, matrix-rich textures, such as those found in lagoonal environments, typically have lower permeability and porosity. The reduced reservoir quality in these settings is partly due to limited interparticle porosity and increased cementation, which restricts fluid movement. Additionally, localized over-pressuring and limited fluid migration may influence porosity distribution in some areas of the Minagish Formation, potentially contributing to the variability in reservoir quality across the field. Further analysis using rock typing and modal analysis will help to clarify these trends and better understand the relationship between depositional factors and reservoir quality.

Minagish Formation in the Magwa field

The Minagish Formation in the Magwa field, part of the greater Burgan area, exhibits a complex depositional model marked by cyclic alternations between low-energy outer ramp (OR) and moderately energetic mid-ramp (MR) environments. It is divided into three members: the Upper Member, Middle Minagish Oolite, and Lower Dense Member. The Upper and Lower Members primarily consist of mudstone and wackestone facies, indicating deep-water, low-energy conditions below the fair-weather wave base, while the Middle Minagish Oolite is made up of packstones and grainstones, reflecting higher energy in the mid-ramp.

These depositional cycles are controlled by relative sea-level fluctuations, with outer ramp facies representing lowstand and transgressive phases, while mid-ramp facies mark a shallowing-upward trend, shaped by storm reworking and grain transport. Core analysis from Magwa wells shows poor reservoir quality, with porosity ranging from 0.5% to 2.8% and permeability between <0.01 mD to 0.2 mD, primarily due to extensive cementation. Reservoir quality is consistently low, with no significant porosity enhancements across lithofacies or depositional environments (Core Laboratories 2016). The formation, predominantly composed of lime wackestone, lime mudstone, and lime packstone, exhibits matrix-dominated textures where porosity development is largely destroyed by cementation. While the Minagish Formation is typically considered to have good to very good reservoir potential in other areas of Kuwait, the reservoir quality is poor due to its matrix-dominated lithofacies, exhibiting low visible and measured porosity and permeability values.

Minagish Formation in Ahmadi Field

The Minagish Formation in the Ahmadi field is interpreted as a homoclinal carbonate ramp system, representing the distal portion of the ramp. The depositional environment is dominated by inner ramp subtidal to mid-ramp conditions, with no well-developed shoals, backshoals, or lagoonal environments. The inner ramp setting is characterized by fine-grained, low-diversity, and argillaceous-rich sediments, primarily composed of wackestones, mudstones, and rare packstones. These deposits are associated with low to moderate energy, often reflecting deposition below the fair-weather wave base, in a low-energy marine environment. The mid-ramp facies, which show evidence of more variable hydrodynamic regimes, contain cleaner, bioturbated wackestones and wacke-packstones, but still lack significant planktonic fauna, reinforcing the inner ramp interpretation (after [Siddiqi et al. 2016](#)). The transition between these two settings is gradual, influenced by storm activity, fluctuations in wave base, and varying sediment input. The cyclical alternation between low-energy, clay-prone deposits and bioturbated, cleaner carbonates suggest a dynamic system with periodic changes in storm intensity and sedimentation rates.

The reservoir quality in the Ahmadi field Minagish Formation is generally very poor to poor, Porosity typically ranges below 16%, while permeability remains predominantly less than 1 mD, owing to the fine-grained, cemented nature of the sediments. The pervasive cementation by calcite and dolomite, along with matrix-supported microporosity, restricts pore space and connectivity. Limited secondary porosity, derived from the dissolution of skeletal grains, offers marginal improvements in reservoir potential. Overall, the Minagish Formation in the Ahmadi area exhibits low reservoir quality, primarily controlled by diagenetic processes and the fine-grained nature of the sediments, with little influence from depositional texture or lithofacies in determining reservoir characteristics.

Minagish reservoir in Umm Gudair Field

The Minagish Formation in the Umm Gudair field exhibits a complex depositional setting characterized by eastward-prograding carbonate ramp deposits, influenced by deltaic progradation towards the end of the Middle Minagish. Ten lithofacies have been identified, consisting of three mud-support facies (MF1-MF3) and seven grain-support facies (MF4-MF10) (after [KISR 2010](#)). The mud-support facies are primarily composed of mudstones and wackestones with bioclasts as the dominant allochems, indicative of low-energy, deeper-water environments. Conversely, the grain-support facies are predominantly packstones and grainstones, often rich in peloids, ooids, and occasionally bioclasts, representing more energetic, shallow-water environments. Notably, small rudist-coral buildups are observed in the Middle Minagish across all studied wells, providing evidence of localized biohermal developments.

The depositional model suggests a transition from outer ramp and mid ramp settings to inner ramp grainshoals and finally to stabilised carbonate sand flats and lagoonal environments. These environments are linked to the eastward progradation of the carbonate ramp, with a marked influence of clastic input from deltaic sources during the late stages of the Middle Minagish. The Middle Minagish is also marked by two main 3rd-order sequence boundaries, with the sedimentary architecture remaining laterally continuous, especially in the lower two-thirds of the interval. Notably, the boundary between the Middle and Upper Minagish is diachronous, with older deposits observed in the western part of the field, closer to the clastic source.

In terms of reservoir characteristics, porosity and permeability are highest in the grainstone and packstone facies, with grain-supported facies exhibiting better pore connectivity compared to the mud-supported facies. Porosity is typically enhanced in areas with minimal cementation and matrix content, while permeability is moderate to high in the grainstone facies, particularly in areas of local biohermal development. However, porosity decreases upwards towards the more argillaceous, fine-grained deposits, limiting reservoir potential in the mudstone and wackestone facies.

Minagish reservoir in Minagish Field

The Minagish reservoir in the Minagish field consists of a variety of depositional facies that span a distal to proximal gradient, ranging from open mid-ramp to lagoonal environments. The **Upper Minagish** primarily consists of fine-grained mud-wackestones, representing an outer ramp environment. These deposits include minor benthic foraminifera, algae, and sponge spicules, indicating moderate-energy, open marine conditions. Porosity within these facies typically ranges from 8-10%, with moderate permeability (1-5 mD), influenced by the fine-grained nature and limited biogenic content. In contrast, the **Shoal** facies is characterized by coarse peloidal and oolitic grainstones and rudstones, indicating very shallow, high-energy environments with substantial marine current activity. These facies exhibit higher porosity (up to 18%) and permeability (10-20 mD), as the coarse-grained textures provide enhanced reservoir potential. Moving proximally, **Back Shoal** facies show finer grains, with peloidal and skeletal packstones, but still maintain moderate to high energy. Porosity and permeability in this interval range from 10-15% and 5-10 mD, respectively (after [Deperiere 2012](#)). **Inner Ramp** facies, composed of bioclastic wackestones and packstones, reflect a low to moderate energy environment, with abundant green algae and benthic foraminifera. These facies exhibit lower porosity (6-9%) and permeability (less than 1 mD) due to their micritic matrix and fine-grained nature. **Lagoonal** facies are dominated by bioclastic mudstones and wackestones, reflecting a very low-energy, restricted lagoonal environment. These deposits show the lowest porosity (3-6%) and permeability (below 0.5 mD) due to significant bioturbation and a clay-rich micritic matrix. Overall, the Minagish reservoir's vertical succession demonstrates a regressive trend from moderate to high-energy environments, offering varying porosity and permeability characteristics across the field.

Depositional Characteristics of the Middle and Upper Minagish in Rugei, Dibdibba Field, and Sawadi Areas

The depositional characteristics of the Upper and Middle Minagish in the Rugei, Dibdibba Field, and Sawadi areas are based on detailed sedimentological data. These deposits reflect a homoclinal carbonate ramp system with distinct facies distribution, influenced by varying energy regimes and detrital input. The Middle Minagish, as observed in the Rugei Area, displays a transition from intertidal (IBD) to marginal/proximal midramp (IME/MP) settings. This is evident from the occurrence of oolitic geobodies, which suggest the development of a marginal shoal corridor and sand bars, with the "Oolitic Minagish" unit marking the base of the Middle Minagish Member (after [Bonin et al. 2016](#)). The shoal corridor, in more marginal ramp positions, is associated with high-energy oolite-rich deposits found in fields like Umm Gudair and Minagish in northern Kuwait.

In contrast, the backshoal oolite-rich deposits (BSH) in the Rugei Area are formed in a more protected environment, where low-energy conditions promote diffuse bioturbation. These oolitic bodies are thought to form elongated patches along the NW-SE shoreline, with a landward decrease in size due to reduced wave energy. The interior inner ramp carbonate sand bars (ISB) are in the uppermost Middle Minagish, with vertical stacking patterns observed. The lateral extent of these sand bars is estimated to be around 0.5-1 km, though precise lateral continuity is challenging to establish. Evidence of marginal shoals in the upper Middle Minagish is scarce in southern Kuwait, possibly due to their migration northward.

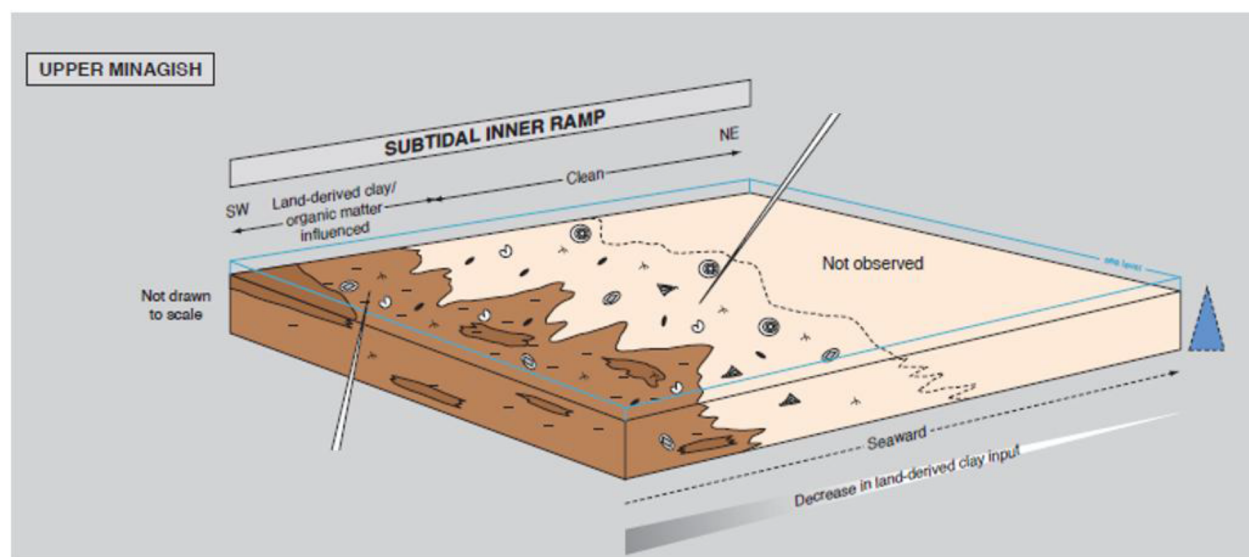


Figure 8—Depositional Model of Upper Minagish in Rugei Field (Bonin et al. 2016).

The Upper Minagish, on the other hand, reflects a more proximal carbonate ramp, with depositional environments characterized by low- and moderate-energy inner ramp deposits influenced by land-derived detrital input. These deposits, especially in the ILE.a and IME.a genetic elements, suggest episodic variations in sediment composition driven by fluvial runoff and autocyclic processes, such as shifts in river courses. The transition from clay/organic matter-prone deposits to cleaner limestones further seaward indicates changes in detrital supply, highlighting the complex interaction between local and regional environmental factors. Both the Middle and Upper Minagish display vertical heterogeneity, which is key to understanding the dynamic depositional history of the area, with variations controlled by factors such as wave energy, fluvial input, and shifts in accommodation space over time.

Minagish reservoir in Rahaiyah Field

The depositional model of the Minagish reservoir in the Rahaiya area, as evidenced by the sedimentological characterization of the Minagish Formation, indicates a setting primarily dominated by deposition on a homoclinal ramp. This ramp is believed to include low-relief shoal complexes and/or paleo highs within the subtidal lagoon. The vertical distribution of sediments suggests a predominance of inner ramp environments, including a backshoal complex, an open-marine subtidal lagoon, and a clay-prone subtidal lagoon. Algal debris found within the clean limestones points to the presence of algal meadows in the shallow lagoon, a characteristic feature of both the Ratawi and Minagish formations. However, there is a slight anomaly in the interpretation of proximal versus distal sub-environments within the inner ramp. Faunal assemblages in the clay-prone intervals, dominated by miliolids and ostracods, imply that factors such as water depth and turbidity may have influenced the ecosystem, as filter-feeding organisms are less likely to thrive in clay-rich environments. Ostracods, however, can tolerate brackish, estuarine conditions, offering insight into the water chemistry and mixing of fluvial and marine waters. (After Green & Phukan, 2017).

Depositional Setting	Lithofacies Association		Lithofacies Composition	Sedimentary Characteristics	Diagenetic Character	Depositional Environment	Lithofacies Distribution
Inner Ramp	Lagoon	Clay-prone Subtidal Lagoon	SL.cl	Wackestone, wacke-packstone and packstone - all argillaceous-rich (a.cl). Marls.	Common argillaceous seams and wisps, low to moderate bioturbation intensity, locally laminated, rarely nodular, restricted faunal content. Miliolid benthic foraminifera and ostracods common along with bivalves within marls.	Lithification by calcite with localised pore-filling cements, minor dolomite and pyrite replacement.	Low-energy subtidal lagoon where clay and clay-sized particles are deposited with variable carbonate content. Clay indicates the increase of terrigenous input into the system.
		Low to moderate Open Marine Subtidal Lagoon	SL	Wackestone, wacke-packstone, packstone, pack-grainstone and floatstone, rare grainstone.	Moderate to common bioturbation, minor argillaceous wisps with peloids, <i>Thalassinoides</i> -rich. Varied faunal assemblage, algal-rich, mainly debris, locally <i>Lithocodium/Bacinnella</i> .	Moderate to strong lithification by calcite cement, localised replacive dolomite and rare replacive pyrite. Pore-lining calcite and dolomite locally common.	Low to moderate-energy subtidal lagoon, open to the sea promoting algal meadows. Clay-prone sediments represent marginal settings to more protected areas.
	Shoal Complex	Backshoal	SH.bs	Packstone, pack-grainstone, grainstone and floatstone.	Grain-rich, medium to coarse grade grains, coated grains are locally common, peloids and skeletal debris with bioturbation. Fauna includes bivalves, echinoids, coarse algal grains and benthic foraminifera.	Moderate to strong lithification by calcite cement and dolomite cement.	Transition between lagoon and shoal (not observed in this core). Grain-rich with large debris well sorted representing the higher energy in the system.

Figure 9—Lithofacies Association in Rahaiya Field (After Green & Phukan, 2017)

These low-energy, clay-rich areas are likely to be more proximal to land, associated with fluvial systems supplying clays that impede carbonate production. Such clay influx, either from fluvial input or along-shore sediment reworking, could alter the water chemistry, further inhibiting carbonate factory processes and restricting carbonate precipitation. Additionally, weak intertidal influences are suggested by sedimentary structures within thin intervals of the core, providing further nuance to the overall depositional model. The reservoir quality evaluation of the Rahaiya field, based on conventional core analysis (CCA) data from the Minagish formations reveals variable porosity and permeability across different lithofacies. While the Minagish Formation exhibits a wider range of porosity from 0.2% to 15.1% and permeability from 0.01 to 34 mD. The majority of the data cluster below 3% porosity and 0.1 mD permeability, with no clear lithofacies trend, suggesting that textural factors do not significantly control reservoir quality (After Green & Phukan, 2017)

Minagish reservoir in Khashman Field

The depositional model of the Minagish reservoir in the Khashman field is dominated by outer and mid-ramp environments, characterized by interbedded calcareous shales, lime mudstones, and wackestones. The **Upper Minagish** consists of outer and mid-ramp deposits, with abundant **Chondrites**-bioturbated lime mudstones and wackestones, and occasional calcareous shales. This section reflects the progradation of mid-ramp facies, with limestone beds likely representing minor progradational cycles where carbonate sediment accumulated in shallower mid-ramp settings. These cycles are interpreted as a result of relative sea-level fluctuations, which allowed the deposition of fine-grained carbonates in a low-energy environment. The **Chondrites** ichnofacies supports this interpretation, indicating a mid-shelf or ramp environment with lower-energy conditions. Rare grainstone laminae within the interval are tempestites, providing a higher-energy signal and representing storm events. These tempestites, though sparse, contribute localized zones of increased permeability. Porosity in the Upper Minagish typically ranges from 3% to 7%, with permeability values generally below 1 mD, reflecting the fine-grained, compacted nature of the sediments.

The **Middle and Lower Minagish** are characterized by similar depositional environments, primarily outer ramp, represented by calcareous shales and interbedded lime mudstones and wackestones. The presence of the **Zoophycos** ichnofacies and common **Chondrites** suggests a similar mid-ramp interpretation, with the progradation of these facies at the tops of minor cycles. Two overall shoaling parasequence sets are identified, with progressively upward-thinning cycles indicative of accommodation space reduction during highstand. These cycles represent a gradual reduction in water depth and deposition

of finer-grained sediments, further reducing porosity and permeability. In the Lower Minagish, porosity typically ranges from 2% to 5%, with permeability values often below 0.1 mD. Diagenetic processes such as cementation, compaction, and dolomitization have further reduced the reservoir quality. Overall, the Minagish reservoir is characterized by low porosity and permeability, primarily influenced by fine-grained, low-energy facies and diagenetic overprints that limit fluid flow potential.

Minagish reservoir in Bahra Field

The Minagish Formation in the Bahra field, located in the northern part of Kuwait, is predominantly characterized as a highstand system tract (HST) deposit, forming part of a prograding ramp system. The depositional model of the Minagish Formation consists of low to moderate-energy environments that transition from outer ramp through to middle ramp settings, reflecting semi-arid climatic conditions (Core Laboratories, 2018). The Lower Minagish interval is primarily composed of lime mudstones with occasional interbeds of lime wackestones. This thick sequence represents deepening-upward cycles that were deposited in a low-energy outer ramp environment, which is often associated with reduced hydrodynamic conditions and restricted water circulation. These facies typically exhibit low porosity and permeability, with porosity values typically less than 5%, and permeability below 1 mD, due to the matrix-supported nature of the deposits and subsequent cementation processes. The Middle Minagish consists of alternating lime mudstones, lime wackestones, and rare dolowackestones, with subordinate dolomitic mudrocks (Core Laboratories, 2018). This sequence reflects an aggradational setting, which may indicate shallowing or coarsening upward cycles, indicative of the transition from outer ramp to lower middle ramp environments during highstand system tract deposition. Porosity in this section is generally low due to cementation but is locally enhanced by moldic, intraparticle, and intercrystal pores, which provide some degree of reservoir potential, albeit at low levels (5-8% porosity and < 2 mD permeability) (Core Laboratories, 2018). Notably, micropores dominate the pore structure, with limited residual oil staining observed in certain intervals, which indicates the presence of hydrocarbons despite low overall reservoir quality. Overall, the Minagish Formation in the Bahra field exhibits limited porosity and permeability, primarily controlled by texture, freshwater flushing, and episodes of subaerial exposure that have resulted in cementation and porosity destruction (Core Laboratories, 2018).

Minagish reservoir in Bihayth Field

The Minagish Formation in the Bihayth field is characterized by well-defined cyclic deposition, comprising 11 parasequences that range from 4 to 12 feet in thickness, reflecting 5th order sea-level fluctuations. Each cycle begins with shale, argillaceous, or darkened grain facies, which overlie an omission or discontinuity surface marked by firm ground at the top of the underlying cycle (Slater and Burgess, 2011). These sediments likely represent transgressive drowning events, deposited under relatively deep, middle to outer ramp or shelf settings below the storm wave base. The increase in clay content suggests the influence of clastic input, which potentially "switches off" the carbonate factory and supports a clastic-dominated environment. The presence of Chondrites colonization further indicates a distal Cruziana ichnofacies, consistent with a dysoxic mid- to outer ramp setting. As the transgressive cycles progressed, the accommodation space gradually filled, leading to the deposition of clean wackestone carbonate facies in relatively shallow, protected lagoonal environments on the middle ramp (Fig.10). These facies, marked by Miliolids and dasyclads, indicate calm, clear, and restricted water conditions. The presence of conical Trocholina suggests subtidal deposition in water depths less than 20 meters, while flattened Trocholina may indicate low-light or turbid conditions, often concentrated along argillaceous seams. These carbonate deposits are capped by bored firm grounds, signaling renewed transgression.

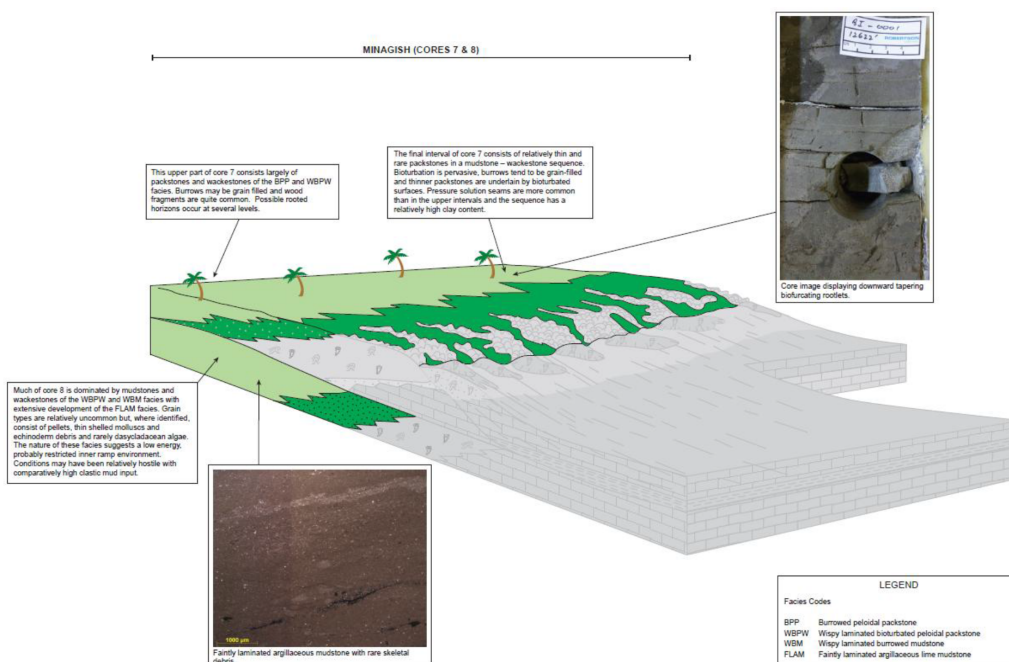


Figure 10—Depositional Model of Minagish reservoir in Bihayth Field (Slater and Burgess, 2011).

The reservoir potential of the Minagish Formation in the Bihayth field is generally poor, particularly within the shale and argillaceous mudstone facies. Reservoir quality improves in areas where dolomitization has occurred but remains limited in skeletal lime mudstone and argillaceous wackestone and packstone due to the presence of abundant matrix. Porosity and permeability are enhanced in clean wackestone and packstone facies, although they are still moderate due to micritic matrix and minor calcite cement, which restricts pore connectivity. Porosity decreases upwards towards discontinuity surfaces, limiting overall reservoir quality. The Minagish reservoir represents storm dominated sedimentation on a homoclinal carbonate ramp. The lower part of (Upper Minagish Member) represents progressive drowning of the inner ramp during renewed transgression with sea level rise. (After Slater and Burgess, 2011).

Table 1—summarizing the Minagish Formation depositional environments, facies, and reservoir quality across different fields in Kuwait (from south to north: Burgan, Magwa, Ahmadi, Khashman, Bahra, and Bihayth fields)

Field	Depositional Environment	Facies	Reservoir Quality
Burgan Field	Progradational oolite-rimmed carbonate ramp; backshoal, mid-ramp, and foreshoal environments; northeastward progradation	Oolitic shoals, peloidal-skeletal shoals, wash-over units, algal and rudist shoals, oncoidal and microbial facies	Porosity: 4%–10% Permeability: 1–10 mD (higher in shoals) Reservoir Quality: Good, especially in oolitic shoals, with significant lateral continuity and progradational stacking of high-energy environments.
Magwa Field	Outer, mid, and inner ramp facies; deep water, distal settings	Fine-grained mudstones, wackestones, lime packstones, bioclasts, peloids	Porosity: 0.5%–2.8% (avg. 1.4%) Permeability: <0.01–0.2 mD (avg. 0.03 mD) Reservoir Quality: Poor, dominated by matrix-supported textures with pervasive cementation. Low porosity and permeability due to cementation
Ahmadi Field	Inner to mid-ramp subtidal to mid-ramp settings; no shoal complexes	Fine-grained low-diversity fauna, argillaceous-rich deposits	Porosity: 4%–8% Permeability: <0.5 mD Reservoir Quality: Poor to moderate; fine-grained facies with some bioturbation, but overall low permeability and porosity due to cementation.
Khashman Field	Outer and mid-ramp environments; Chondrites and Zoophycos ichnofacies indicative of low-energy distal settings	Lime mudstones, wackestones, fine-grained calcareous shales	Porosity: 2%–10% Permeability: <1 mD Reservoir Quality: Generally poor to moderate; low permeability and porosity in fine-grained calcareous facies. cementation and compaction dominate the overall reservoir quality. Diagenetic processes significantly reduce permeability.

Field	Depositional Environment	Facies	Reservoir Quality
Bahra Field	Outer Ramp low to moderate-energy, prograding ramp system	Lime mudstones, lime wackestones, packstones, and rare dolomitic mudrocks	Porosity: 5%–8% Permeability: <2 mD Reservoir Quality: Poor to moderate; reduced by cementation, but localized improvements due to moldic, interparticle, and intercrystalline pores. Minor residual oil staining indicates hydrocarbons despite overall low reservoir potential.
Bihayth Field	Outer Ramp Transgressive carbonate ramp system; moderate to low energy with storm-dominated deposition	Shale, argillaceous grain facies, clean wackestone, Trocholina, bioturbated zones, Miliolids,	Porosity: 3%–6% Permeability: <0.5 mD Reservoir Quality: Poor to moderate; low permeability in shales and argillaceous mudstones but improved in clean wackestone and packstone facies. Cementation and matrix textures dominate but some localized reservoir potential in cleaner intervals.

Table 2—summarizing the depositional environments, facies, and reservoir quality of the Minagish Formation across several fields from East to West (Burgan to Umm Gudair, Minagish, Rugei, Dibdibba, and Sawadi fields), including Reservoir quality

Field	Depositional Environment	Facies	Porosity (%)	Permeability (mD)	Reservoir Quality
Burgan Field	Progradational carbonate ramp (Oolite-rimmed, Mid-ramp, Outer ramp)	Oolitic shoals, peloidal-skeletal shoals, algal, rudist, microbial, backshoal, lagoonal	6-18%	1-20 mD	High lateral facies heterogeneity, varying reservoir quality with better potential in coarse-grained facies.
Umm Gudair Field	Eastward-prograding carbonate ramp (Outer ramp to Inner ramp)	Mudstones, wackestones, packstones, grainstones, rudist-coral buildups, bioherms	5-15%	1-10 mD	High porosity in grain-supported facies, lower porosity in mud-supported facies. Moderate to high permeability in grainstone facies.
Minagish Field	Distal to proximal carbonate ramp (Outer ramp to Lagoonal)	Mud-wackestones, peloidal and oolitic grainstones, skeletal packstones, bioclastic wackestones	3-18%	<0.5 - 20 mD	Porosity and permeability increase in coarse-grained, shoal facies. Fine-grained facies exhibit poor reservoir quality.
Rugei Field	Homoclinal carbonate ramp with marginal/proximal mid-ramp and backshoal	Oolitic geobodies, backshoal oolite-rich, inner ramp carbonate sand bars, sand bars	8-15%	1-10 mD	Good reservoir potential in oolitic deposits and carbonate sand bars, lower potential in finer-grained backshoal facies.
Dibdibba Field	Homoclinal carbonate ramp (Mid-ramp, Inner ramp)	Oolitic, peloidal grainstones, backshoal, inner ramp carbonate sand bars	8-12%	1-12 mD	Moderate to high reservoir quality in oolitic and grainstone facies, poorer in more clay-dominant facies.
Sawadi Field	Proximal carbonate ramp with land-derived detrital input (Fluvial influence)	Mudstones, wackestones, grainstones, clay/organic matter-rich facies	6-10%	<1 mD	Low reservoir quality in fine-grained facies, moderate in cleaner limestones.

Minagish Formation depositional environments, facies, and reservoir quality across different fields South to North

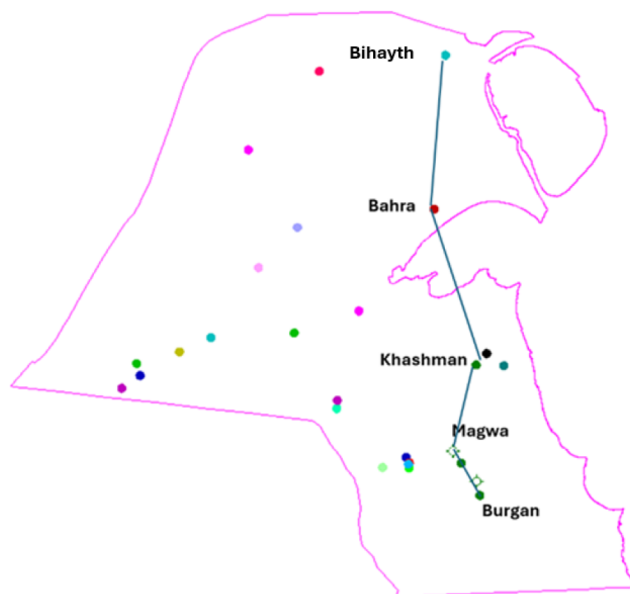


Figure 11—Maps showing the North-South correlation of Minagish reservoir with Adjacent fields

The Minagish Formation across various fields in Kuwait exhibits significant variation in depositional environments, facies types, and reservoir quality. In the Burgan Field, a progradational oolite-rimmed carbonate ramp features high-energy shoal complexes (oolitic, peloidal-skeletal, and rudist), resulting in good reservoir quality, with porosity of 4%–10% and permeability up to 10 mD, especially within oolitic shoals. In contrast, the Magwa Field, characterized by distal outer to inner ramp facies and fine-grained sediments (mudstones, wackestones), shows poor reservoir quality, with average porosity of 1.4% and permeability around 0.03 mD, mainly due to pervasive cementation (Fig. 12). Similarly, the Ahmadi Field, with no shoal complexes and dominated by fine-grained, argillaceous facies, offers poor to moderate quality, marked by low permeability (<0.5 mD) despite slightly higher porosity (4%–8%). The Khashman Field represents low-energy outer ramp settings with ichnofossil evidence of distal conditions. It consists mostly of mudstones and calcareous shales, yielding poor to moderate reservoir quality, constrained by porosity up to 10% but very low permeability (<1 mD). In the Bahra Field, a low to moderate-energy outer ramp system exhibits facies such as lime mudstones and packstones, with poor to moderate reservoir characteristics, due to cementation, although localized improvements exist via moldic and interparticle porosity. Lastly, the Bihayth Field, deposited in a transgressive carbonate ramp system with storm influence, shows poor to moderate quality, where shales and argillaceous facies limit reservoir performance, but clean wackestones offer some potential. Overall, reservoir quality declines northward, from the high-energy shoal-dominated Burgan Field to the fine-grained, cementation-dominated northern fields.

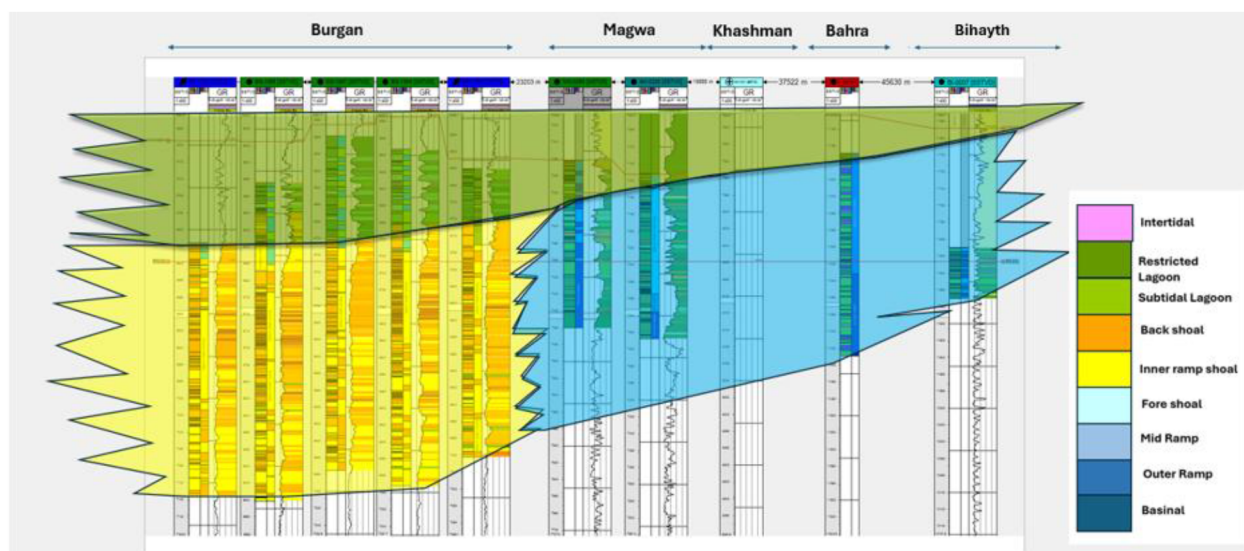


Figure 12—North South Correlation of Minagish Reservoir with adjacent fields

Minagish Formation depositional environments, facies, and reservoir quality across different fields West to East

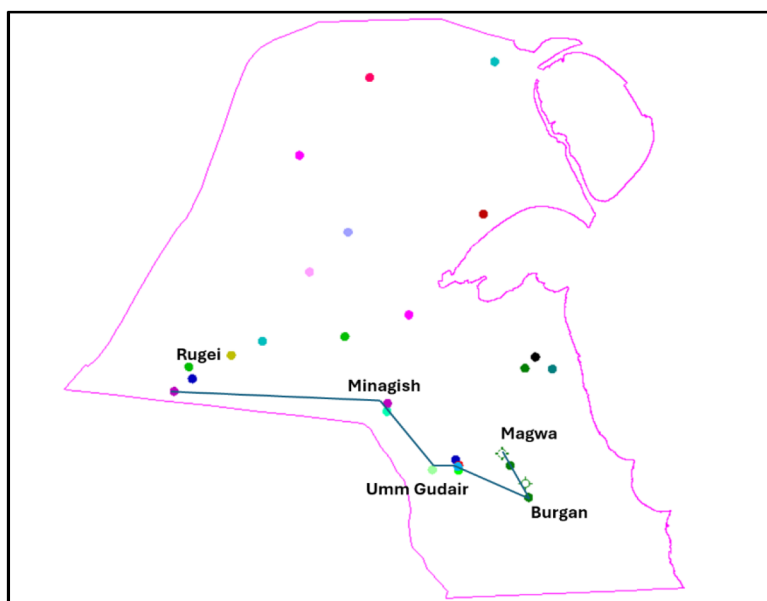


Figure 13—Map showing the East West correlation of Minagish reservoir with Adjacent fields

The Minagish Formation across different fields West to East demonstrates a range of depositional settings and reservoir qualities influenced by facies distribution and texture (Fig.14). The Burgan Field, characterized by a progradational carbonate ramp, contains oolitic, skeletal, and microbial facies with porosity ranging from 6–18% and permeability up to 20 mD, offering good reservoir quality in coarser-grained units. In the Umm Gudair Field, an eastward-prograding ramp system hosts a mix of grain-supported and mud-supported facies, with porosity of 5–15% and permeability of 1–10 mD, showing higher quality in grainstones and rudist buildups. The Minagish Field spans outer ramp to lagoonal settings, where coarse-grained shoal facies (oolitic and skeletal) exhibit porosity up to 18% and permeability reaching 20 mD, while finer-grained mudstones have poor reservoir properties. The Rugei Field, a homoclinal ramp with mid-ramp to backshoal facies, shows porosity of 8–15% and permeability of 1–10 mD, with the best quality in

oolitic geobodies and sand bars. Similarly, the Dibdibba Field presents moderate to high reservoir potential in oolitic and grainstone facies, with porosity between 8–12% and permeability of 1–12 mD, although clay-rich intervals reduce quality. In contrast, the Sawadi Field, influenced by fluvial input, contains organic-rich and fine-grained facies, leading to lower reservoir quality with porosity of 6–10% and permeability below 1 mD, though cleaner limestones show moderate potential. Overall, reservoir quality across the fields correlates strongly with facies type, with higher quality in grain-supported, coarser facies and significantly reduced potential in fine-grained or cemented intervals.

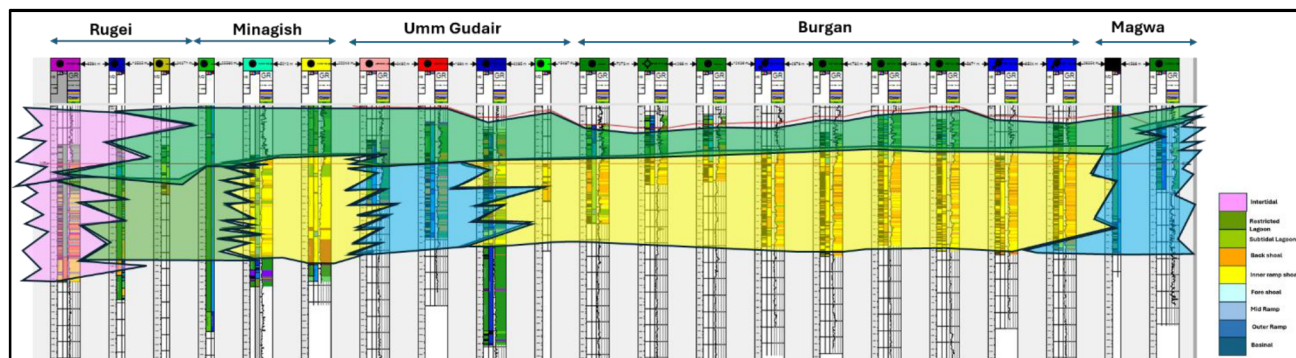


Figure 14—East West Correlation of Minagish reservoir with adjacent fields

Results and Discussions

The study underscores the pronounced east-west and north-south depositional controls on the Minagish Formation across Kuwait, with clear correlations that significantly influence reservoir qualities. From east to west, the Minagish oolitic facies are most well-developed in the Umm Gudair and Minagish fields, with high porosity and permeability, rendering them prolific hydrocarbon producers. These high-energy grain-supported shoal deposits facilitate excellent reservoir connectivity and facilitate efficient hydrocarbon extraction. However, moving westward towards Rugei and Sawadi fields, the depositional environment transitions into intertidal, low-energy settings characterized by finer-grained sediments, increased cementation, and argillaceous content, which markedly reduce reservoir quality. Northward from Burgan, the depositional continuum shifts from high-energy, oolitic shoals in the south and east to lower-energy, distal deposits in Magwa, Ahmadi, Khashman, Bahra, and Bihayth fields (Fig.15). These areas exhibit coarser facies with comparatively poor reservoir properties due to pervasive cementation and reduced porosity, resulting in a more heterogeneous, compartmentalized reservoir system with limited fluid flow pathways. The variations along both axes profoundly impact the heterogeneity, fluid flow, and hydrocarbon distribution within the formation.

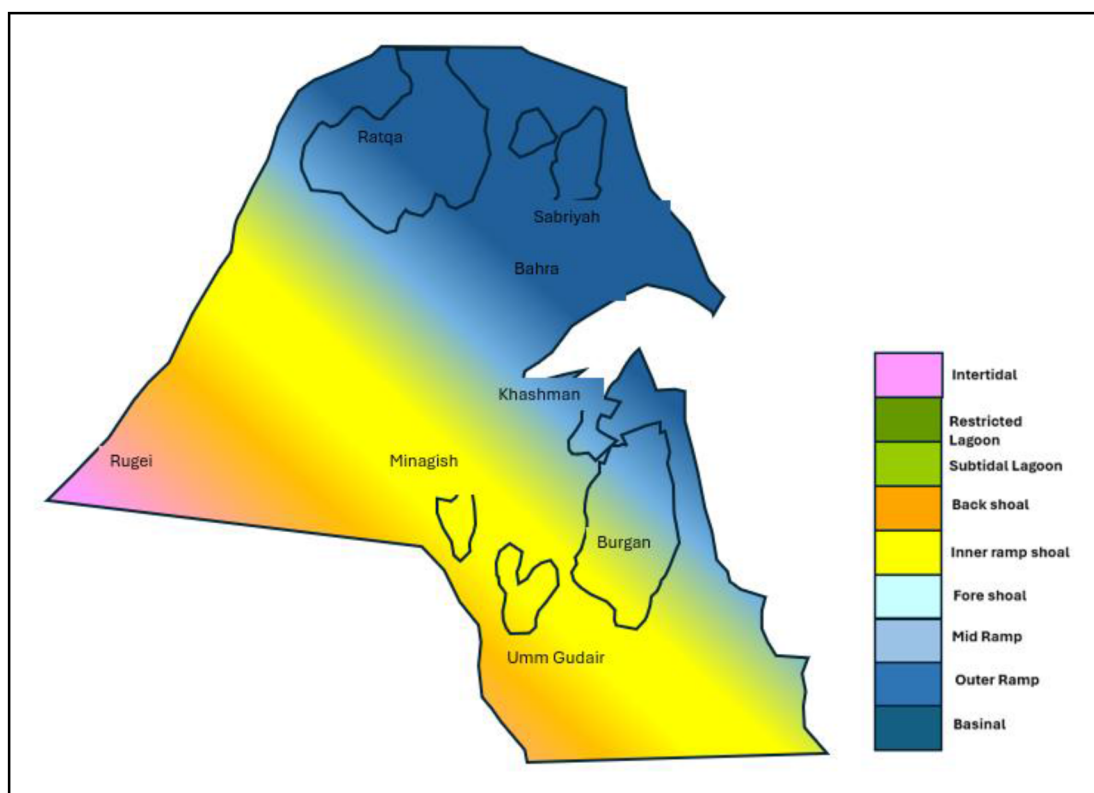


Figure 15—Showing the Depositional Environments of Minagish reservoir along different fields in Kuwait

An important consideration in this context is the potential connectivity of the Burgan Minagish reservoir with the adjacent fields based on facies distribution. Two depositional scenarios are possible: one where the grainstone shoals of the inner ramp environment in Burgan, Umm Gudair, and Minagish fields were deposited on isolated structural highs, implying that these reservoirs may not be laterally connected; and another model in which the grainstone shoals are laterally continuous across these fields, suggesting potential aquifer connectivity. However, well data from the area between Burgan and Umm Gudair primarily shows outer ramp facies, which contradicts the possible continuity of the grainstone shoal. Interestingly, historical field pressure trends hint at a degree of aquifer communication between these fields. Further evaluation using additional sedimentological data and reservoir engineering analyses, such as interference testing between the two fields, could provide greater clarity on the extent and nature of this connectivity.

Within the Burgan field, the Minagish reservoir shows significant facies variation that influences reservoir management and production strategies substantially. The upper Minagish, consisting mainly of clay-prone, cemented mudstones and marls, exhibits poor reservoir quality, often necessitating stimulation and hydraulic fracturing to facilitate hydrocarbons flow. The high permeability associated with oolitic shoal facies in the high-energy zones promotes rapid water encroachment during water flooding due to the nearly unity Kv/Kh ratio, which indicates a highly homogeneous reservoir. This homogeneity results in quick vertical water breakthrough, thereby reducing sweep efficiency and complicating reservoir management under high drawdown conditions. Further, the inner ramp facies deteriorate in the northern fields such as Magwa, Ahmadi, Khashman, Bahra, and Bihayth, where depositional environments shift towards finer, low-porosity, and low-permeability sediments. Eastward in the Umm Gudair and Minagish fields, Oolitic grainstone is well developed resulting in proven hydrocarbon production, whereas further west, in the Rugei field it transitions into intertidal sand flats and tidal settings.

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